

***Generation Interconnection
Revised System Impact Study Report
For
Queue Request
AC2-186
“Harrington 25 kV”***

September 2020

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Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The Interconnection Customer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Freepoint Solar LLC, the Interconnection Customer (IC), has proposed solar generating project AC2-186, a 10 MW MFO (3.8 MW Capacity) facility to be located in Kent County, Delaware. PJM studied AC2-186 as an injection into the Delmarva Power & Light Company (DPL) system at the Harrington 69 kV Substation and evaluated it for compliance with reliability criteria for summer peak conditions in 2020.

The planned in-service date, as requested by the IC during the project kick-off call, was December 2018. This date is not attainable due to additional PJM studies and the Transmission Owner's construction schedule.

Point of Interconnection

The Interconnection Customer requested that queue projects AC2-186, AC2-187, and AC2-188 utilize the same Point of Interconnection which will be a new terminal position at the Delmarva Power & Light Harrington 69 kV Substation (see Attachment 1).

Transmission Owner Scope of Work

Substation Interconnection Estimate

Scope: Construct a new terminal position at the Harrington 69 kV Substation.

Estimate: \$ 2,086,000

Construction Time: 24-36 months

Major Equipment Included in Estimate:

• Power Circuit Breaker, 69 kV, 2000A, 40kA, 3 cycle	Qty. 1
• Disconnect Switch, 69 kV, 2000A, Manual Wormgear, Arcing Horns	Qty. 4
• CT/VT Combination Units, 69 kV	Qty. 3
• Disconnect Switch Stand, High, 69 kV, Steel	Qty. 4
• Disconnect Switch Stand, Low, 69 kV, Steel	Qty. 2
• CT/VT Stand, Single Phase, Low, 69 kV, Steel	Qty. 3
• CVT Stand, Single Phase, Low, 69 kV, Steel	Qty. 3
• Relay Panel, Transmission Line, FL/BU (20")	Qty. 1
• Control Panel, 69 kV Circuit Breaker (10")	Qty. 1
• Bus Support Structure, 3 phase, 69 kV, Steel	Qty. 3
• Take-off structure, 69 kV	Qty. 1

Estimate Assumptions:

- The existing Harrington substation property will not need to be expanded. The cost of purchasing any new property is not included in the conceptual estimate.
- No new storm water management is required.
- Clearing and grading of the existing site is minimal.
- Permitting will be performed by DPL.
- Existing control house is adequate for expansion.

Required Relaying and Communications

New protection relays are required for the new line terminals. An SEL-421 will be required for primary protection and an SEL-311C will be required for back-up protection. A 20" relay panel will be required for each transmission line (2 total).

New protection relays are required for the new generator terminal. An SEL-487 will be required for primary protection and an SEL-387 will be required for back-up protection. A 20" relay panel is required.

A SEL-451 relay on a 10" breaker control panel will be required for the control and operation of the new 69 kV circuit breakers (4 total).

The cost of the required relay and communications is included in the Substation Interconnection Estimate.

Metering

Three phase 69 kV revenue metering points will need to be established. DPL will purchase and install all metering instrument transformers as well as construct a metering structure. The secondary wiring connections at the instrument transformers will be completed by DPL's metering technicians. The metering control cable and meter cabinets will be supplied and installed by DPL. DPL will install conduit for the control cable between the instrument transformers and the metering enclosure. The location of the metering enclosure will be determined in the construction phase. DPL will provide both the Primary and the Backup meters. DPL's meter technicians will program and install the Primary & Backup solid state multi-function meters for each new metering position. Each meter will be equipped with load profile, telemetry, and DNP outputs. The IC will be provided with one meter DNP output for each meter. DPL will own the metering equipment for the interconnection point, unless the IC asserts its right to install, own, and operate the metering system.

The Interconnection Customer will be required to make provisions for a voice quality phone line within approximately 3 feet of each Company metering position to facilitate remote interrogation and data collection.

It is the IC's responsibility to send the data that PJM and DPL requires directly to PJM. The IC will grant permission for PJM to send DPL the following telemetry that the IC sends to PJM: real time MW, MVAR, volts, amperes, generator status, and interval MWH and MVARH.

The estimate for DPL to design, purchase, and install metering as specified in the aforementioned scope for metering is included in the Substation Interconnection Estimate.

Interconnection Customer Scope of Direct Connection Work

The Interconnection Customer is responsible for all design and construction related activities on their side of the Point of Interconnection. Site preparation, including grading and an access road, as necessary, is assumed to be by the IC. Route selection, line design, and right-of-way acquisition of the direct connect facilities is not included in this report, and is the responsibility of the IC. Protective relaying and metering design and installation must comply with DPL's applicable standards. The IC is also required to provide revenue metering and real-time telemetering data to PJM in conformance with the requirements contained in PJM Manuals M-01 and M-14 and the PJM Tariff.

DPL Interconnection Customer Scope of Direct Connection Work Requirements:

- DPL requires that an IC circuit breaker is located within 500 feet of the DPL substation to facilitate the relay protection scheme between DPL and the IC at the Point of Interconnection (POI).

Special Operating Requirements

1. DPL will require the capability to remotely disconnect the generator from the grid by communication from its System Operations facility. Such disconnection may be facilitated by a generator breaker, or other method depending upon the specific circumstances and the evaluation by DPL.
2. DPL reserves the right to charge the Interconnection Customer operation and maintenance expenses to maintain the Interconnection Customer attachment facilities, including metering and telecommunications facilities, owned by DPL.

Network Impacts

The Queue Project AC2-186 was evaluated as a 10.0 MW (Capacity 3.8 MW) injection at the Harrington 69 kV substation in the DPL area. Project AC2-186 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC2-186 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

None

Short Circuit

(Summary of impacted circuit breakers)

No issues identified

Stability and Reactive Power Requirement

(Results of the dynamic studies should be inserted here)

No issues identified. See Attachment 2 for full report.

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

Light Load Analysis - 2020

Light Load Studies to be conducted during the System Impact Study phase (as required by PJM Manual 14B).

Not applicable

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

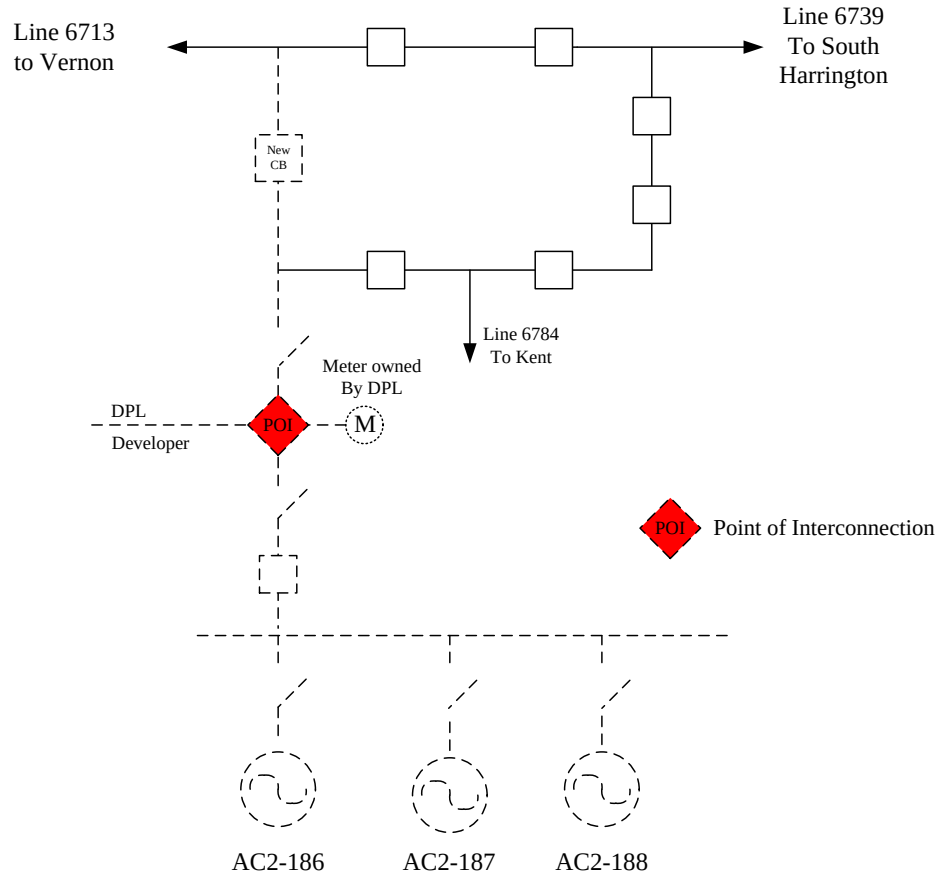
None

Delmarva Power and Light Costs

Cost estimates will further be refined as a part of the Facilities Study for this project. The Interconnection Customer will be responsible for all costs incurred by DPL in connection with the AC2-186 project. Such costs may include, but are not limited to, any transmission system assets currently in DPL's rate base that are prematurely retired due to the AC2-186 project. PJM shall work with DPL to identify these retirement costs and any additional expenses. DPL reserves the right to reassess issues presented in this document and, upon appropriate justification, submit additional costs related to the AC2-186 project.

Attachment 1 – Single Line Diagram

AC2 – 186, 187, 188 Harrington 69 kV New 69 kV Terminal Position



If location of generator is greater than 500 feet from substation, additional developer circuit breaker will be required.

Attachment 2 – Dynamic Simulation Analysis

AC2-186/187/188 System Impact Study

Dynamic Simulation Analysis

Version 0

Executive Summary

Generator Interconnection Requests AC2-186/187/188 are for a combined 50 MW Maximum Facility Output (MFO) solar powered generating facility with a Point of Interconnection (POI) at the existing Harrington 69kV Substation in the Delmarva Power and Light transmission system, Kent County, Delaware.

This report describes a dynamic simulation analysis of AC2-186/187/188 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2020 Summer Peak case, modified to include applicable queue projects. AC2-186/187/188 has been dispatched online at maximum power output, with POI voltage of (1.014 p.u.), consistent with the default generator reference voltage specified in PJM Manual 03 Transmission Operations Section 3.3.3 for generator connections to the PJM 69 kV system.

The AC2-186/187/188 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 20-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are covered by P1 and P4 contingencies.

The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

No relevant High Speed Reclosing (HSR) contingencies were identified.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the remaining fault contingencies tested on the 2020 Summer Peak case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AC2-186/187/188 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).

- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.

1. Introduction

Generator Interconnection Requests AC2-186/187/188 are for a combined 50 MW Maximum Facility Output (MFO) solar powered generating facility with a Point of Interconnection (POI) at the existing Harrington 69kV Substation in the Delmarva Power and Light transmission system, Kent County, Delaware.

This analysis is effectively a screening study to determine whether the addition of AC2-186/187/188 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AC2-186/187/188 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

2. Description of Project

Generator Interconnection Requests AC2-186/187/188 are for a combined 50 MW Maximum Facility Output (MFO) solar powered generating facility with a Point of Interconnection (POI) at the existing Harrington 69kV Substation in the Delmarva Power and Light transmission system, Kent County, Delaware.

Figure 1 shows the simplified one-line diagram of the AC2-186/187/188 load flow models. Table 1 through to Table 3 list the parameters given in the impact study data and the corresponding parameters of the AC2-186/187/188 load flow models.

The dynamic models for the AC2-186/187/188 plant are based on the SMASC_C131_33_IVF111 PSS/E version 33 user defined model supplied by the Developer in the attachments to the System Impact Study (SIS) Data Sheets.

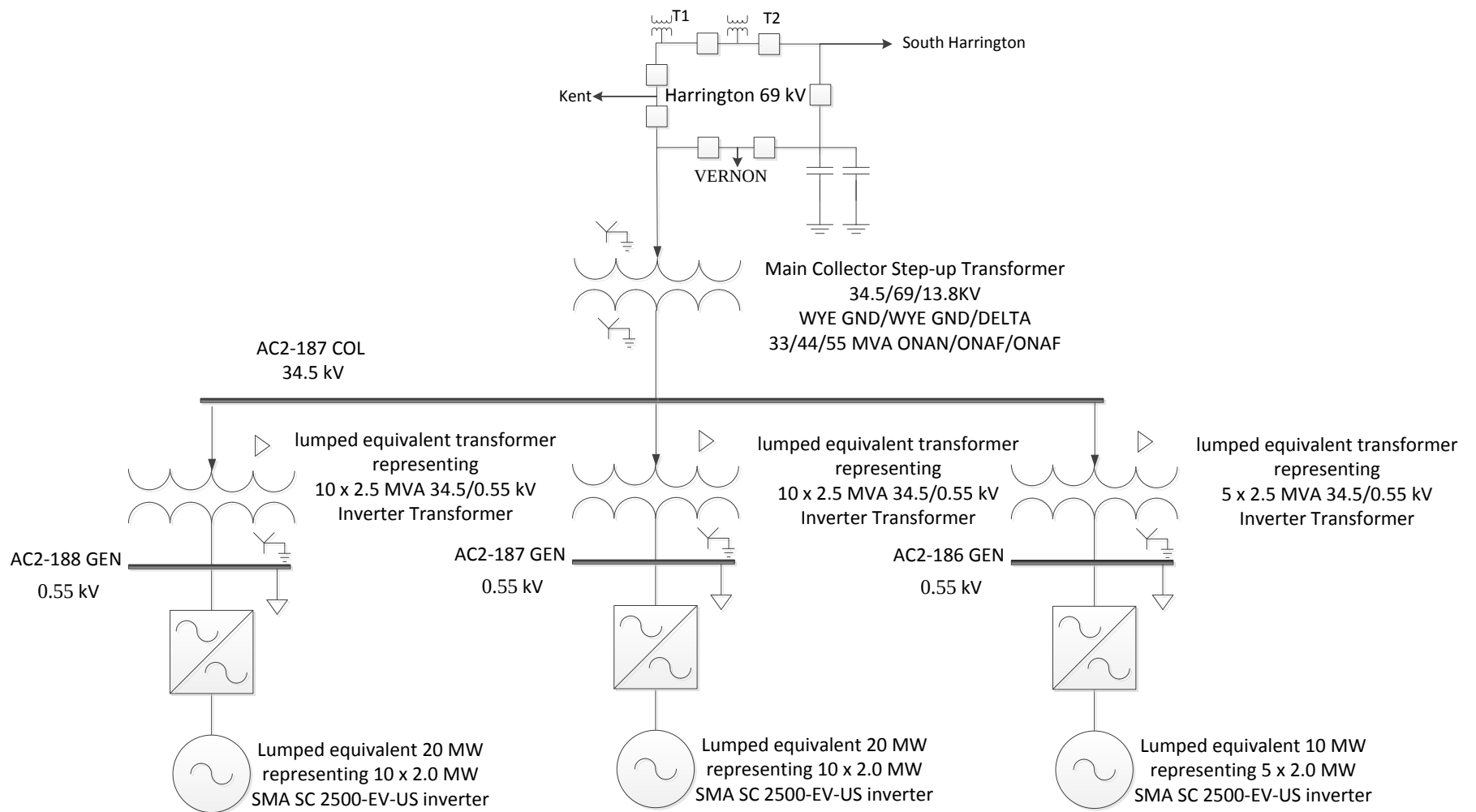


Figure 1: AC2-186/187/188 Plant Model

Table 1: AC2-186 Plant Model

	Impact Study Data	Model
Inverters	5 × 2.0 MW SC 2500-EV-US PV Inverters MVA base = 2.5 MVA Terminal voltage = 0.55 kV	Lumped equivalent representing 10 × 2.0 MW SC 2500-EV-US Inverters Pgen 10.194 MW Pmax 10.194 MW Pmin 0 MW Qmax 7 MVA_r Qmin -7 MVA_r Mbase 12.5 MVA Zsorce 0.0 + j10000 pu @Mbase
GSU transformer	5 x 34.5/0.55 kV two winding transformers Transformer MVA base = 2.5 MVA Rating = 2.5 MVA Impedance = 0.0045 + j0.067 pu @ MVA base Number of taps = N/A Tap step size = N/A	Lumped equivalent representing 5 x 34.5/0.55 kV two winding transformers Transformer MVA base = 12.5 MVA Rating = 12.5 MVA Impedance = 0.0045 + j0.067 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
Collector step-up transformer (AC2-186/187/188)	69/34.5 kV two winding transformer Rating = 33/44/55 MVA (OA/F1/F2) Transformer MVA base = 33 MVA Impedance = 0.0019 + j0.09 pu @ MVA base Number of taps = Not Provided Tap step size = Not Provided	69/34.5 kV two winding transformer Rating = 33/44/55 MVA (OA/F1/F2) Transformer MVA base = 33 MVA Impedance = 0.0019 + j0.09 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
Auxiliary load	N/A	Not modeled
Station load	0.02 MW + 0.013 MVA _r	0.02 MW + 0.013 MVA _r modeled at the inverter bus
Collector System Equivalent	Impedance = 0.015385+ j 0.011364 Suseptance = 0.001670	Impedance = 0.015385+ j 0.011364 Suseptance = 0.001670
Transmission line (AC2-186/187/188)	Length = 0.25 miles Impedance = 0.00144+ j 0.0042 Suseptance = 0.00006	Length = 0.25 miles Impedance = 0.00144+ j 0.0042 Suseptance = 0.00006

Table 2: AC2-187 Plant Model

	Impact Study Data	Model
Inverters	10 × 2.0 MW SC 2500-EV-US PV Inverters MVA base = 2.5 MVA Terminal voltage = 0.55 kV	Lumped equivalent representing 10 × 2.0 MW SC 2500-EV-US Inverters Pgen 20.52 MW Pmax 20.52 MW Pmin 0 MW Qmax 14 MVar Qmin -14 MVar Mbase 25.0 MVA Zsorce 0.0 + j10000 pu @Mbase
GSU transformer	10 x 34.5/0.55 kV two winding transformers Transformer MVA base = 2.5 MVA Rating = 2.5 MVA Impedance = 0.0045 + j0.067 pu @ MVA base Number of taps = N/A Tap step size = N/A	Lumped equivalent representing 10 x 34.5/0.55 kV two winding transformers Transformer MVA base = 25 MVA Rating = 25 MVA Impedance = 0.0045 + j0.067 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
Collector step-up transformer (AC2-186/187/188)	69/34.5 kV two winding transformer Rating = 33/44/55 MVA (OA/F1/F2) Transformer MVA base = 33 MVA Impedance = 0.0019 + j0.09 pu @ MVA base Number of taps = Not Provided Tap step size = Not Provided	69/34.5 kV two winding transformer Rating = 33/44/55 MVA (OA/F1/F2) Transformer MVA base = 33 MVA Impedance = 0.0019 + j0.09 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
Auxiliary load	N/A	Not modeled
Station load	0.04 MW + 0.013 MVar	0.04 MW + 0.013 MVar modeled at the inverter bus
Collector System Equivalent	Impedance = 0.003660+ j 0.001530 Suseptance = 0.002300	Impedance = 0.003660+ j 0.001530 Suseptance = 0.002300
Transmission line (AC2-186/187/188)	Length = 0.25 miles Impedance = 0.00144+ j 0.0042 Suseptance = 0.00006	Length = 0.25 miles Impedance = 0.00144+ j 0.0042 Suseptance = 0.00006

Table 3: AC2-188 Plant Model

	Impact Study Data	Model
Inverters	10 × 2.0 MW SC 2500-EV-US PV Inverters MVA base = 2.5 MVA Terminal voltage = 0.55 kV	Lumped equivalent representing 10 × 2.0 MW SC 2500-EV-US Inverters Pgen 20.35 MW Pmax 20.35 MW Pmin 0 MW Qmax 14 MVar Qmin -14 MVar Mbase 25.0 MVA Zsorce 0.0 + j10000 pu @Mbase
GSU transformer	10 x 34.5/0.55 kV two winding transformers Transformer MVA base = 2.5 MVA Rating = 2.5 MVA Impedance = 0.0045 + j0.067 pu @ MVA base Number of taps = N/A Tap step size = N/A	Lumped equivalent representing 10 x 34.5/0.55 kV two winding transformers Transformer MVA base = 25 MVA Rating = 25 MVA Impedance = 0.0045 + j0.067 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
Collector step-up transformer (AC2-186/187/188)	69/34.5 kV two winding transformer Rating = 33/44/55 MVA (OA/F1/F2) Transformer MVA base = 33 MVA Impedance = 0.0019 + j0.09 pu @ MVA base Number of taps = Not Provided Tap step size = Not Provided	69/34.5 kV two winding transformer Rating = 33/44/55 MVA (OA/F1/F2) Transformer MVA base = 33 MVA Impedance = 0.0019 + j0.09 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
Auxiliary load	N/A	Not modeled
Station load	0.04 MW + 0.013 MVar	0.04 MW + 0.013 MVar modeled at the inverter bus
Collector System Equivalent	Impedance = 0.005+ j 0.0041 Suseptance = 0.003650	Impedance = 0.005+ j 0.0041 Suseptance = 0.003650
Transmission line (AC2-186/187/188)	Length = 0.25 miles Impedance = 0.00144+ j 0.0042 Suseptance = 0.00006	Length = 0.25 miles Impedance = 0.00144+ j 0.0042 Suseptance = 0.00006

3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process¹.

The selected load flow scenario is the RTEP 2020 Summer Peak case with the following modifications:

- a) Addition of all applicable queue projects prior to AC2-186/187/188.
- b) Addition of AC2-186/187/188 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AC2-186/187/188.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.

The AC2-186/187/188 initial conditions are listed in **Table** , indicating maximum power output, with AC2-186/187/188 regulating POI voltage of (1.014 p.u.), consistent with the default generator reference voltage specified in PJM Manual 03 Transmission Operations Section 3.3.3 for generator connections to the PJM 69 kV system.

Table 4: AC2-186/187/188 machine initial conditions

Bus	Name	Unit	PGEN (MW)	QGEN (MVAR)	ETERM (p.u.)	POI Voltage (p.u.)
932644	AC2-186 GEN 0.5500	1	10.194	2.1	1.03	1.014
932654	AC2-187 GEN 0.5500	1	20.52	2.1	1.03	1.014
932664	AC2-188 GEN 0.5500	1	20.35	2.1	1.03	1.014

Generation within the vicinity of AC2-186/187/188 has been dispatched online at maximum output (PMAX). The dispatch of generation in the vicinity of AC2-186/187/188 is given in Attachment 3.

¹ Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

4. Fault Cases

Tables 5 listed the contingencies and results that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are covered by P1 and P4 contingencies.

The system was tested for a system intact condition and the fault types listed above. No relevant High Speed Reclosing (HSR) contingencies were studied.

5. Evaluation Criteria

This study is focused on AC2-186/187/188, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) The system with AC2-186/187/188 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 3%.
- b) The AC2-186/187/188 is able to ride through faults (except for faults where protective action trips AC2-186/187/188).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 4, with results summarized in Table 5.

The frequency protection was disabled due to the PSSE deficiency in calculating frequencies for 3ph fault at POIs.

For the fault contingencies tested in this study:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AC2-186/187/188 generator was able to withstand all contingencies.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

7. Mitigations

No Mitigations were found to be required.

Table 5: Fault list

P0: Steady State

Fault ID	Duration
P0.00	Steady State 20 sec run

P1: Three Phase Faults with normal clearing

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P1.00	3ph @ AB2-135 POI – Chestertown 69kV line, normal clear	9	Stable
P1.01	3ph @ Chestertown – McCleans – Lynch – Kennedyville – Massey – Church 69kV line, normal clear	9	Stable
P1.02	3ph @ Chestertown – Clough – Church 69kV line, normal clear	9	Stable
P1.03	3ph @ AB2-036 – Oil City – Steele 138kV line, normal clear	9	Stable
P1.04	3ph @ Church 69/138kV Tx #1, normal clear	9	Stable
P1.05	3ph @ Church 69/138kV Tx #2, normal clear	9	Stable
P1.06	3ph @ Church – AB2-135 POI, normal clear	9	Stable
P1.07	3ph @ Church – I.B. Corners – AC1-220 69kV line, normal clear	9	Stable
P1.08	3ph @ Centerville – Wye Mills 69kV line, normal clear	9	Stable
P1.09	3ph @ AB2-135 POI – New Meredith – Kent 69kV line, normal clear	9	Stable
P1.10	3ph @ Kent – Cheswold 69kV line, normal clear	9	Stable
P1.11	3ph @ Kent – NRG Dover 69kV line, normal clear loss of NRG G1 and G2	9	Stable
P1.12	3ph @ Kent – Vaughn – Harrington 69kV line, normal clear loss of Vaughn unit	9	Stable
P1.13	3ph @ Harrington – AC2-186/187/188 69kV line, normal clear loss of AC2-186/187/188	9	Stable
P1.14	3ph @ Harrington – Vaughn – Kent 69kV line, normal clear loss of Vaughn unit	9	Stable
P1.15	3ph @ Harrington 69/25kV TF 1, normal clear	9	Stable
P1.16	3ph @ Harrington 69/25kV TF 2, normal clear	9	Stable

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P1.17	3ph @ Harrington – South Harrington 69kV line, normal clear	9	Stable
P1.18	3ph @ Harrington capacitors, normal clear	9	Stable
P1.19	3ph @ Harrington – Vernon 69kV line, normal clear	9	Stable
P1.20	3ph @ South Harrington – Kratz – Bridgeville 69kV, normal clear loss of South Harrington capbank	9	Stable
P1.21	3ph @ South Harrington 69/138kV TF, normal clear	9	Stable
P1.22	3ph @ South Harrington – Harrington 69kV line, normal clear	9	Stable

P4: SLG Stuck Breaker (SB) Faults at Backup Clearing

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.01	SLG @ Chestertown 69/25kV Tx #1, SB @ Chestertown, delayed clear loss of Chestertown – McCleans – Lynch – Kennedeyville – Massey – Church 69kV line	9 / 22	Stable
P4.02	SLG @ Chestertown – McCleans – Lynch – Kennedeyville – Massey – Church 69kV line, normal clear, SB @ Chestertown, delayed clear loss of AB2-133	9 / 22	Stable
P4.03	SLG @ Chestertown 69/25kV Tx #2, SB @ Chestertown, delayed clear loss of Chestertown – Clough – Church 69kV line	9 / 22	Stable
P4.04	SLG @ Chestertown – Clough – Church 69kV line, SB @ Chestertown, delayed clear loss of AB2-133	9 / 22	Stable
P4.05	SLG @ Chestertown 69/25 kV Tx #1, SB @ Chestertown, delayed clear loss of Chestertown 69/25kV Tx #2	9 / 22	Stable
P4.06	SLG @ Church – AB2-036 138kV line, SB @ Church, delayed clear loss of Church – AB1-141 138kV line.	9 / 21	Stable
P4.07	SLG at Church 69 kV on Massey – Kennedeyville – Lynch – McCleans – Chestertown circuit 6727 normal clear. Breaker 7210 stuck. Fault cleared with loss of Church 69 kV / 138 kV transformer AT2.	9 / 22	Stable

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.08	SLG at Church 69 kV on Massey – Kennedyville – Lynch – McCleans – Chestertown circuit 6727, normal clear. Breaker 7290 stuck. Fault cleared with loss of Church – AB2-135 POI circuit 6704.	9 / 22	Stable
P4.09	SLG at Church 69 kV on Clough – Chestertown circuit 6773. Breaker 7240 stuck. Fault cleared with loss of I.B. Corners – Price circuit 6710.	9 / 22	Stable
P4.10	SLG at Church 69 kV on Clough – Chestertown circuit 6773. Breaker 7260 stuck. Fault cleared with loss of Church 69 kV / 138 kV Transformer AT1.	9 / 22	Stable
P4.11	SLG at Church 69 kV on AB2-135 POI circuit 6704. Breaker 7290 stuck. Fault cleared with loss of Church 69 kV on Massey – Kennedyville – Lynch – McCleans – Chestertown circuit 6727.	9 / 22	Stable
P4.12	SLG at Church 69 kV on AB2-135 POI circuit 6704. Breaker 7220 stuck. Fault cleared with loss of Church 69 kV / 25 kV Transformer T3 and Church 69 kV / 25 kV Transformer T4, Z1-081.	9 / 22	Stable
P4.13	SLG at Church 69 kV on I.B. Corners – Price circuit 6710. Breaker 7250 stuck. Fault cleared with loss of Church 69 kV / 25 kV Transformer T3 and Church 69 kV / 25 kV Transformer T4, Z1-081.	9 / 22	Stable
P4.14	SLG at Church 69 kV on I.B. Corners – Price circuit 6710. Breaker 7240 stuck. Fault cleared with loss of Clough – Chestertown circuit 6773.	9 / 22	Stable
P4.15	SLG at Church 69 kV on 69 kV / 138 kV transformer AT1. Breaker 60 stuck. Fault cleared with loss of Church 69 kV / 138 kV Transformer AT2.	9 / 22	Stable
P4.16	SLG at Church 69 kV on 69 kV / 138 kV transformer AT1. Breaker 7260 stuck. Fault cleared with loss of Clough – Chestertown circuit 6773.	9 / 22	Stable
P4.17	SLG at Church 69 kV on 69 kV / 138 kV transformer AT2. Breaker 60 stuck. Fault cleared with loss of Church 69 kV / 138 kV Transformer AT1.	9 / 22	Stable
P4.18	SLG at Church 69 kV on 69 kV / 138 kV transformer AT2. Breaker 7210 stuck. Fault cleared with loss of Church 69 kV on Massey – Kennedyville – Lynch – McCleans – Chestertown circuit 6727.	9 / 22	Stable
P4.19	SLG @ Kent – Cheswold 69kV line, Stuck Breaker @ Kent, delayed clear loss of Kent – New Meredith – AB2-135 POI 69kV line	9 / 22	Stable
P4.20	SLG @ Kent – NRG Dover 69kV line, normal clear loss of NRG Dover units, Stuck Breaker @ Kent, delayed clear loss of Kent – New Meredith – AB2-135 POI 69kV line	9 / 22	Stable
P4.21	SLG @ Kent – Cheswold 69kV line, Stuck Breaker @ Kent, delayed clear loss of Kent – Scuse (Not Modeled)	9 / 22	Stable

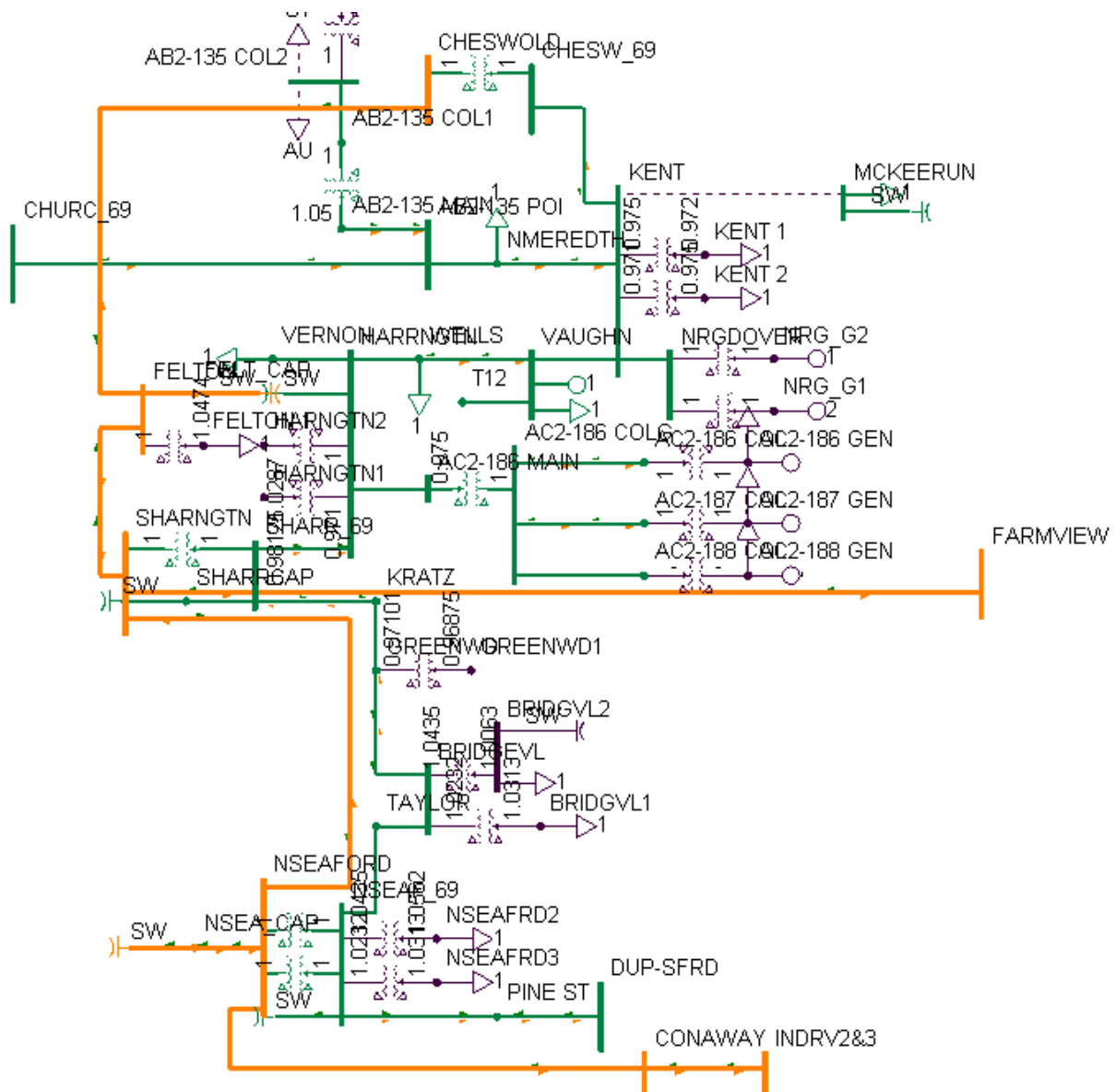
Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.22	SLG @ Kent 69/25kV Tx T2, Stuck Breaker @ Kent 69kV, delayed clear loss of NRG Dover units	9 / 22	Stable
P4.23	SLG @ Kent 69/25kV Tx T2, Stuck Breaker @ Kent 69kV, delayed clear loss of Kent – Vaughn – Harrington 69kV line, Vaughn unit.	9 / 22	Stable
P4.24	SLG @ Harrington 69kV, normal clear loss of Harrington – Vaughn – Kent 69 kV, SB @ Harrington 69kV, delayed clear loss of AC2-186/187/188	9 / 22	Stable
P4.25	SLG @ Harrington 69kV, normal clear loss of Harrington – Vaughn – Kent 69 kV, SB @ Harrington 69kV, delayed clear loss of Harrington 69/25kV TF 1	9 / 22	Stable
P4.26	SLG @ Harrington 69kV, normal clear loss of Harrington 69/25kV TF 2, SB @ Harrington 69kV, delayed clear loss of Harrington 69/25kV TF 1	9 / 22	Stable
P4.27	SLG @ Harrington 69kV, normal clear loss of Harrington 69/25kV TF 2, SB @ Harrington 69kV, delayed clear loss of Harrington – South Harrington 69kV line	9 / 22	Stable
P4.28	SLG @ Harrington 69kV, normal clear loss of Harrington 69kV capbank, SB @ Harrington 69kV, delayed clear loss of Harrington – South Harrington 69kV line	9 / 22	Stable
P4.29	SLG @ Harrington 69kV, normal clear loss of Harrington 69kV capbank, SB @ Harrington 69kV, delayed clear loss of Harrington – Vernon 69kV line	9 / 22	Stable
P4.30	SLG @ Harrington 69kV, normal clear loss of AC2-186/187/188, SB @ Harrington 69kV, delayed clear loss of Harrington – Vernon 69kV line	9 / 22	Stable
P4.31	SLG @ South Harrington 69kV, normal clear loss of South Harrington 138/69kV TF, SB @ South Harrington 69kV, delayed clear loss of South Harrington capbank, South Harrington – Kratz – Bridgeville 69kV line	9 / 22	Stable
P4.32	SLG @ South Harrington 69kV, normal clear loss of South Harrington 138/69kV TF, SB @ South Harrington 69kV, delayed clear loss of South Harrington – Harrington 69kV line	9 / 22	Stable
P4.33	SLG @ South Harrington 69kV, normal clear loss of South Harrington capbank, South Harrington – Kratz – Bridgeville 69kV line, SB @ South Harrington 69kV, delayed clear loss of South Harrington – Harrington 69kV line	9 / 22	Stable

P5: SLG Fault with Delayed (Zone 2) Clearing

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P5.01	SLG at 80% of 138kV line from AB2-036 to Oil City – Steele, delayed clear.	9 / 37	Stable
P5.02	SLG at 80% of 69 kV line from Chestertown to Clough – Church circuit 6773. Delayed clearing at Chestertown.	9 / 42	Stable
P5.03	SLG at 80% of 69 kV line from Church to Massey – Kennedyville – Lynch – Chestertown circuit 6727. Delayed clearing at Church.	9 / 42	Stable
P5.04	SLG at 80% of 69 kV line from Church to Clough – Chestertown circuit 6773. Delayed clearing at Church.	9 / 42	Stable
P5.05	SLG at 80% of 69 kV line from Church on AB2-135 POI circuit 6704. Delayed clearing at Church.	9 / 42	Stable
P5.06	SLG at 80% of 69 kV line from Church on I.B. Corners – AC1-220 circuit 6710. Delayed clearing at Church.	9 / 42	Stable
P5.07	SLG at 80% of 69kV line from Chestertown – Lynch – Kennedyville – Massey – Church, Delayed clearing at Chestertown.	9/42	Stable
P5.08	SLG at Church 69/138kV transformer AT1, Delayed clearing	/42	Stable
P5.09	SLG at 80% of 69 kV line from AB2-135 POI on Church circuit 6704. Delayed clearing at AB2-135 POI.	9 / 42	Stable
P5.10	SLG at 80% of 69 kV line from AB2-135 POI on New Meredith - Kent circuit 6704. Delayed clearing at AB2-135 POI.	9 / 42	Stable
P5.11	SLG at 80% of 69 kV line from Kent on New Meredith - AB2-135 POI circuit 6704. Delayed clearing at Kent.	9 / 42	Stable
P5.12	SLG @ 80% of 69kV line from Kent on Cheswold, delayed clearing at Kent	9 / 42	Stable
P5.13	SLG @ 80% of 69kV line from Kent on Vaughn – Harrington, delayed clearing at Kent, loss of Vaughn unit	9 / 42	Stable
P5.14	SLG @ 80% of 69kV line from Kent on NRG Dover, delayed clearing at Kent, loss of NRG Dover units.	9 / 42	Stable
P5.15	SLG @ 80% of Harrington – Vaughn – Kent 69kV, delayed clearing @ Harrington, loss of Vaughn unit	9 / 42	Stable
P5.16	SLG @ 80% of Harrington – South Harrington 69kV, delayed clearing @ Harrington	9 / 42	Stable
P5.17	SLG @ 80% of South Harrington – Harrington 69kV, delayed clearing @ South Harrington	9 / 42	Stable

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P5.18	SLG @ 80% of South Harrington – Kratz - Bridgeville 69kV, delayed clearing @ South Harrington	9 / 42	Stable

Attachment 1. PSS/E Model One Line Diagram



Attachment 2. AC2-186/187/188 PSS/E Dynamic Model

/AC2-186 + AC2-187 + AC2-188

932644 'USRMDL' 1 'SMAPPC18' 4 0 4 28 9 21 932641 232217 1 0

1 0.5 0.25 0.04 0.0 0.1 1.0 0.04 0

0 1 0 10 -0.001

0.1 1.0 0.04 0 1 0.2 0.2

0.8 0.91 0.915 1.060 1.055 0.3 1.0/

/ CtlMod KP_PF KI_PF PFXdcrTm PFDB KP_Vol KI_Vol VolXdcrTm VolDB

/ PFNomTot VolNomTot QNomTot VolDroop HybCtlTun

/ KP_P KI_P PXdcrTm PDB PNomTot QCommTm PCommTm

/ PFLim FRTThrhLo1 FRTThrhLo2 FRTThrhHi1 FRTThrhHi2 FRTHldTm

Reserved

932644 'USRMDL' 1 'SMASC131' 1 1 0 76 15 193

1.0 1.0 0.0 0.8 1.0 0.0 1.0 1.0 0.35

1.0 0.0 5.0 0.5 2.0

0.0 0.9 0.5 1.0 0.9 0.9

0.0 0.2 0.05 0.4

0.35 0.35 1.0 2.0 2.0 2.0 0.1 0.098 0.1 0.098 0.2 0.2

0.01 1.0 20.0 0.5 0.61 0.62 0.0 0.125 0.1

1.2 0.1 1.18 1.0 1.15 2.0 0.88 12.0 0.6 5.0 0.5 3.0

65.0 0.1 64.0 1.0 61.5 3.0 59.3 5.0 57.0 3.0 50.0 0.1

30.0 0.0 10.0 30.0 0.0 1.0 1.0/

/ PPrim PWNom QVArNom PFLIM PFPF PFPFExt QVArMod QoDEna

QoDQMax

/ VArCtlVol_Volref VArCtlVol_VolDB VArCtlVol_VArGra

VArCtlVol_VArMax VArCtlVol_VArTm

/ PFPFExtStr PFPFStr PFWStr PFPFExtStop PFPFStop PFWStop

/ WCtlHzMod PHzStr PHzStop PWGra

/ WGra VArGra DGSMOD DGSArGraNom DGSArGraNomHi

DGSArGraNomLo DbVolNomMax DbVolNomMaxH DbVolNomMin DbVolNomMinH

DGSqRcvrTm DGSNqRcvrTm

/ VArCmdFltTm FRTPreErrVena FRTPreErrTm FRTSwOffTm FRTThrshld1

FRTThrshld2 VCtlCharEna VCtlCharTm VCtlCorTm

/ VCtlMax VCtlMaxTm VCtlhhLim VCtlhhLimTm VCtlhLim VCtlhLimTm

VCtlLim VCtlLimTm VCtlllLim VCtlllLimTm VCtlMin VCtlMinTm

/ HzCtlMax HzCtlMaxTm HzCtlhhLim HzCtlhhLimTm HzCtlhLim

HzCtlhLimTm HzCtlllLim HzCtlllLimTm HzCtlllLimTm HzCtlllLimTm HzCtlMin HzCtlMinTm

/ KPLL1 PLLFlag KPPLL2 KIPLL2 Reserved Reserved GenTrpFlag

932654 'USRMDL' 1 'SMAPPC18' 4 0 4 28 9 21 932641 232217 1 0

1 0.5 0.25 0.04 0.0 0.1 1.0 0.04 0

0 1 0 10 -0.001

0.1 1.0 0.04 0 1 0.2 0.2

0.8 0.91 0.915 1.060 1.055 0.3 1.0/

/ CtlMod KP_PF KI_PF PFXdcrTm PFDB KP_Vol KI_Vol VolXdcrTm VolDB

```

/          PFNomTot VolNomTot QNomTot VolDroop HybCtlTun
/          KP_P KI_P PXdcrTm PDB PNomTot QCommTm PCommTm
/          PFLim FRTThrhLo1 FRTThrhLo2 FRTThrhHi1 FRTThrhHi2 FRTHldTm
Reserved
932654 'USRMDL' 1 'SMASC131' 1 1 0 76 15 193
          1.0 1.0 0.0 0.8 1.0 0.0 1.0 1.0 0.35
          1.0 0.0 5.0 0.5 2.0
          0.0 0.9 0.5 1.0 0.9 0.9
          0.0 0.2 0.05 0.4
          0.35 0.35 1.0 2.0 2.0 2.0 0.1 0.098 0.1 0.098 0.2 0.2
          0.01 1.0 20.0 0.5 0.61 0.62 0.0 0.125 0.1
          1.2 0.1 1.18 1.0 1.15 2.0 0.88 12.0 0.6 5.0 0.5 3.0
          65.0 0.1 64.0 1.0 61.5 3.0 59.3 5.0 57.0 3.0 50.0 0.1
          30.0 0.0 10.0 30.0 0.0 1.0 1.0/
/          PPrim PWNom QVArNom PFLIM PFPF PFPFExt QVArMod QoDEna
QoDQMax
/          VArCtlVol_Volref VArCtlVol_VolDB VArCtlVol_VArGra
VArCtlVol_VArMax VArCtlVol_VArTm
/          PFPFExtStr PFPFStr PFWStr PFPFExtStop PFPFStop PFWStop
/          WCtlHzMod PHzStr PHzStop PWGra
/          WGra VArGra DGSMOD DGSArGraNom DGSArGraNomHi
DGSArGraNomLo DbVolNomMax DbVolNomMaxH DbVolNomMin DbVolNomMinH
DGSqRcvrTm DGSNqRcvrTm
/          VArCmdFltTm FRTPreErrVena FRTPreErrTm FRTSwOffTm FRTThrshld1
FRTThrshld2 VCtlCharEna VCtlCharTm VCtlCorTm
/          VCtlMax VCtlMaxTm VCtlhhLim VCtlhhLimTm VCtlhLim VCtlhLimTm
VCtlLim VCtlLimTm VCtlLimTm VCtlLimTm VCtlMin VCtlMinTm
/          HzCtlMax HzCtlMaxTm HzCtlhhLim HzCtlhhLimTm HzCtlhLim
HzCtlhLimTm HzCtlLim HzCtlLimTm HzCtlLimTm HzCtlLimTm HzCtlMin HzCtlMinTm
/          KPLL1 PLLFlag KPPLL2 KIPLL2 Reserved Reserved GenTrpFlag

```

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932664 'USRMDL' 1 'SMAPPC18' 4 0 4 28 9 21 932641 232217 1 0
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          0 1 0 10 -0.001
          0.1 1.0 0.04 0 1 0.2 0.2
          0.8 0.91 0.915 1.060 1.055 0.3 1.0/
/          CtlMod KP_PF KI_PF PFXdcrTm PFDB KP_Vol KI_Vol VolXdcrTm VolDB
/          PFNomTot VolNomTot QNomTot VolDroop HybCtlTun
/          KP_P KI_P PXdcrTm PDB PNomTot QCommTm PCommTm
/          PFLim FRTThrhLo1 FRTThrhLo2 FRTThrhHi1 FRTThrhHi2 FRTHldTm
Reserved
932664 'USRMDL' 1 'SMASC131' 1 1 0 76 15 193
          1.0 1.0 0.0 0.8 1.0 0.0 1.0 1.0 0.35
          1.0 0.0 5.0 0.5 2.0
          0.0 0.9 0.5 1.0 0.9 0.9

```

0.0 0.2 0.05 0.4
 0.35 0.35 1.0 2.0 2.0 2.0 0.1 0.098 0.1 0.098 0.2 0.2
 0.01 1.0 20.0 0.5 0.61 0.62 0.0 0.125 0.1
 1.2 0.1 1.18 1.0 1.15 2.0 0.88 12.0 0.6 5.0 0.5 3.0
 65.0 0.1 64.0 1.0 61.5 3.0 59.3 5.0 57.0 3.0 50.0 0.1
 30.0 0.0 10.0 30.0 0.0 1.0 1.0/
 / PPrim PWNom QVArNom PFLIM PFPF PFPFExt QVArMod QoDEna
 QoDQMax
 / VArCtlVol_Volref VArCtlVol_VolDB VArCtlVol_VArGra
 VArCtlVol_VArMax VArCtlVol_VArTm
 / PFPFExtStr PFPFStr PFWStr PFPFExtStop PFPFStop PFWStop
 / WCtlHzMod PHzStr PHzStop PWGra
 / WGra VArGra DGSMOD DGSArGraNom DGSArGraNomHi
 DGSArGraNomLo DbVolNomMax DbVolNomMaxH DbVolNomMin DbVolNomMinH
 DGSqRcvrTm DGSNqRcvrTm
 / VArCmdFltTm FRTPreErrVEna FRTPreErrTm FRTSwOffTm FRTThrshld1
 FRTThrshld2 VCtlCharEna VCtlCharTm VCtlCorTm
 / VCtlMax VCtlMaxTm VCtlhhLim VCtlhhLimTm VCtlhLim VCtlhLimTm
 VCtlLim VCtlLimTm VCtlLim VCtlLimTm VCtlMin VCtlMinTm
 / HzCtlMax HzCtlMaxTm HzCtlhhLim HzCtlhhLimTm HzCtlhLim
 HzCtlhLimTm HzCtlLim HzCtlLimTm HzCtlLim HzCtlLimTm HzCtlMin HzCtlMinTm
 / KPLL1 PLLFlag KPPLL2 KIPLL2 Reserved Reserved GenTrpFlag

Attachment 3. AC2-186/187/188 PSS/E Case Dispatch

Bus Number	Bus Name	Id	In Service	PGen (MW)	PMax (MW)	PMin (MW)	QGen (Mvar)	QMax (Mvar)	QMin (Mvar)
230927	CDRCRK_REAC 230.00	1	0	0	0	0	-20	0	-20
232124	138TH ST 138.00	1	1	0	0	0	-33.69	75	-69
232616	GEN FOOD 13.200	1	1	15.2	15.2	0	0	1	0
232632	IR SVC 16.000	1	1	0	0	0	-71.99	150	-150
232813	VAUGHN 69.000	1	1	3	3	0	0	0	0
232851	DUP-SFR1 12.470	1	1	10	10	0	0	0	0
232901	NORTHST 69.000	1	1	52	52	5	0	15.6	0
232904	IR4 26.000	4	1	414.2	414.2	0	-2.93	4.9	-2.93
232907	VN8 18.000	8	1	153	153	0	9.088	28	-15
232910	NRG_G1 13.800	2	1	44	44	0	0.545	27	-20
232911	NRG_G2 13.800	1	1	44	44	0	0.545	27	-20
232920	IR10 13.200	1	1	16.1	16.1	0	0	15	0
232922	MR3 13.000	3	1	102	102	35	0	35	0
232923	MR1 12.500	1	1	17	17	6	0	12	0
232924	MR2 12.500	2	1	17	17	6	0	12	0
232925	NEL SVC 16.000	1	1	0	0	0	0.329	150	-150
901171	SMYRNA 2 13.800	1	1	53	53	30	0.396	28	-18
901172	SMYRNA 1 13.800	1	1	48	48	30	0.067	25	0
916281	Z1-081 C 24.900	1	1	2.28	2.28	0	0	0	0
924824	AB2-135 GEN 0.6000	1	1	65.4	65.4	0	15.36	21.5	-21.5
926721	AC1-154 C 69.000	1	1	1.2	1.2	0	0.396	0.396	-0.396
927321	AC1-229 C 138.00	1	1	3.8	3.8	0	1.254	1.254	-1.254
927322	AC1-229 E 138.00	1	1	6.2	6.2	0	-0.487	1.254	-1.254
932644	AC2-186 GEN 0.5500	1	1	10.19	10.19	0	-1.954	3.3	-3.3
932654	AC2-187 GEN 0.5500	1	1	20.52	20.52	0	-1.954	6.6	-6.6
932664	AC2-188 GEN 0.5500	1	1	20.35	20.35	0	-1.954	6.6	-6.6